

GORONTALO JALALUDDIN, Indonesia

WMO: 970480

Lat: 0.6385N

Lon: 122.8524E

Elev: 2

StdP: 101.30

Time Zone: 8.00 (E08)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.8	22.4	17.6	12.6	28.9	18.6	13.5	28.5	8.5	30.9	7.8	30.5	0.1	140	0.307

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	7.5	33.7	25.0	33.2	25.1	32.7	25.1	26.9	31.3	26.6	30.9	26.4	30.6	4.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.7	21.0	28.5	25.5	20.7	28.3	25.2	20.3	28.1	84.7	31.5	83.4	31.3	82.3	31.0	30.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.4	6.0	5.2	18.1	35.4	2.8	1.1	16.1	36.2	14.4	36.8	12.8	37.4	10.8	38.2
			18.1	28.9	2.1	1.0	16.6	29.7	15.4	30.3	14.3	30.9	12.8	31.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.7	27.4	27.4	27.8	28.1	28.0	27.5	27.2	27.3	27.6	28.0	28.0	27.8
	DBStd	0.89	0.80	0.80	0.84	0.84	0.85	0.92	0.84	0.82	0.83	0.80	0.81	0.79
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6451	539	488	552	544	558	523	532	536	528	558	541	552
	CDD18.3	3409	281	254	293	294	299	273	273	278	278	300	291	294
	CDH23.3	33234	2636	2392	2908	2914	2984	2543	2544	2741	2867	3019	2852	2834
	CDH26.7	11759	824	769	1025	1028	1053	834	851	1008	1184	1214	1022	947
(14) Wind	WSAvg	1.4	1.3	1.4	1.4	1.2	1.0	1.1	1.6	2.1	1.7	1.3	1.1	1.1
(15) Precipitation	PrecAvg	1817	197	141	158	169	183	166	124	74	80	134	194	195
	PrecMax	2777	502	355	489	316	456	339	271	292	348	329	502	429
	PrecMin	1108	84	35	24	46	58	20	19	16	22	10	106	40
	PrecStd	440	79	91	115	53	87	88	79	72	72	96	87	92
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.7	32.7	33.6	33.9	33.2	32.7	32.7	33.1	34.1	34.3	34.1	33.3
		MCWB	26.0	25.4	25.4	25.8	26.2	26.0	25.6	24.6	24.3	24.4	25.1	26.1
	2%	DB	31.7	31.9	32.6	32.9	32.6	32.0	31.9	32.4	33.4	33.5	33.1	32.3
		MCWB	25.7	25.4	25.3	25.7	26.1	25.7	25.2	24.4	24.2	24.5	25.4	26.0
	5%	DB	31.2	31.3	31.9	32.1	32.0	31.4	31.3	31.9	32.7	32.8	32.2	31.6
		MCWB	25.6	25.3	25.2	25.7	25.9	25.6	25.0	24.2	24.1	24.6	25.6	25.9
	10%	DB	30.5	30.6	31.1	31.3	31.3	30.7	30.7	31.1	31.9	32.0	31.3	30.9
		MCWB	25.4	25.2	25.2	25.7	25.8	25.5	24.9	24.2	24.1	24.8	25.6	25.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.7	26.5	26.6	27.1	27.3	27.1	26.7	26.1	26.5	26.7	26.9	27.1
		MCDB	31.0	30.9	31.2	31.6	31.7	31.2	31.1	30.7	31.5	31.2	31.6	31.3
	2%	WB	26.2	26.1	26.1	26.6	26.7	26.6	26.1	25.6	25.8	26.2	26.4	26.6
		MCDB	30.4	30.5	30.6	30.9	30.9	30.6	30.4	30.2	30.6	30.8	30.9	30.8
	5%	WB	25.9	25.7	25.7	26.2	26.5	26.2	25.7	25.2	25.4	25.8	26.1	26.2
		MCDB	30.0	29.8	29.9	30.4	30.4	30.1	29.7	29.6	30.0	30.3	30.4	30.2
	10%	WB	25.6	25.4	25.5	25.9	26.1	25.8	25.3	24.7	25.0	25.5	25.8	25.9
		MCDB	29.3	29.2	29.4	29.9	29.8	29.4	29.1	29.0	29.4	29.6	29.7	29.8
(35) Mean Daily Temperature Range	5% DB	MDBR	6.8	7.1	7.5	7.5	7.4	7.2	7.4	8.1	9.3	8.9	7.8	7.1
		MCDBR	7.6	7.9	8.3	8.2	8.2	8.1	8.5	9.2	10.4	10.1	8.8	7.9
		MCWBR	2.5	2.5	2.5	2.4	2.6	2.8	2.8	2.8	3.1	3.0	2.6	2.5
	5% WB	MCDBR	7.2	7.3	7.5	7.6	7.6	7.3	7.5	7.9	8.7	8.4	7.8	7.4
		MCWBR	2.7	2.6	2.5	2.6	2.7	2.8	2.9	2.8	3.0	2.8	2.7	2.6
(40) Clear-Sky Solar Irradiance	taub		0.426	0.428	0.438	0.435	0.425	0.428	0.424	0.440	0.455	0.503	0.457	0.435
	taud		2.438	2.426	2.394	2.402	2.446	2.453	2.468	2.403	2.347	2.190	2.343	2.421
	Ebn at Noon		904	905	890	872	858	842	852	856	861	828	868	890
	Edh at Noon		119	122	126	121	112	109	109	120	130	153	130	120
(44) All-Sky Solar Radiation	RadAvg		5.08	5.34	5.57	5.58	5.31	4.81	5.08	5.45	5.88	5.74	5.38	5.08
	RadStd		0.24	0.34	0.43	0.28	0.21	0.28	0.46	0.39	0.40	0.26	0.25	0.32

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page